Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction

Hassan Waqas Saleem, Rahat Zeeshan Saleem, and Hina Maryam

***Abstract*—A conformal prediction region promises to contain a vehicle's true future path a fixed fraction of the time. We ask what happens to that promise when a trajectory forecaster is calibrated on one driving dataset and deployed, without new labels, on another. Using AutoBot-Ego forecasters calibrated on Argoverse 2 and evaluated on nuScenes and on Waymo Open Motion, and the reverse direction with nuScenes as the source, we measure a coverage gap of up to 23 percentage points at a 90% nominal level. The gap is confirmed on models trained to match published state-of-the-art accuracy and is pre-registered on the third dataset before it was touched. Stronger point prediction makes the gap larger, not smaller. Reweighting the calibration set by an estimated covariate-shift likelihood ratio makes it worse, and a label-free normalised score only partially repairs it. We give a training-conditional guarantee that holds even when labels arrive scene by scene rather than one agent at a time, the realistic case for driving data, where a naive per-agent guarantee is shown to fail outright. With 30 to 40 labelled target scenes it restores coverage close to nominal at roughly 1.5 to 2 times the size of an oracle region, a result that replicates across both transfer directions and all three datasets. We also localise the failure to turning manoeuvres and left-hand traffic, and show that several measured input-level differences between the datasets explain only a minority of it. The results argue that conformal guarantees for trajectory prediction should not be assumed to transfer across datasets without an explicit, budgeted recalibration step.**

***Index Terms*—Conformal prediction, trajectory prediction, distribution shift, uncertainty quantification, autonomous driving.**

AUTONOMOUS vehicles plan around forecasts of what nearby road users will do. A forecast is only as useful as the statement it makes about its own reliability: a planner that knows a predicted region contains the true future with probability at least $1 - \alpha$ can keep clear of that region and of nothing larger. Conformal prediction (CP) supplies such a statement for any black-box predictor, with a finite-sample guarantee that requires nothing beyond exchangeability of calibration and test data [1], [2], and it has been adopted for trajectory prediction and safe planning [3]–[5].

Exchangeability is exactly what deployment breaks. A forecaster is calibrated on the data at hand—one set of cities, one sensor suite, one annotation pipeline—and then driven somewhere else. That accuracy drops when a trajectory predictor moves between datasets is documented [6], and that predictive uncertainty degrades under dataset shift is well known for classifiers [7]. What is not known is what happens to the *guarantee itself* when a conformally calibrated forecaster is moved, without any new labels, to a different real driving dataset: whether the stated $1 - \alpha$ survives, how badly it fails, and whether the standard remedies for shift repair it.

The conformal literature offers several remedies, each resting on an assumption. Weighted CP restores validity under covariate shift when the likelihood ratio is known or can be estimated [8]; more general results bound the coverage loss by a distance between calibration and test distributions [9]; online methods track shift by feeding realised errors back into the threshold [10], [11], an idea carried to trajectory prediction [5] and to robotic settings more broadly [12]; and a generative-prior approach has been proposed for planning under environment shift [13]. What these do not establish is the *static, label-free, dataset-to-dataset* setting in which a forecaster is frozen, calibrated once, and deployed: how large the failure is on real data, whether the covariate-shift remedy suffices, and how many labelled target scenes it takes to detect or repair the failure.

Contributions: (i) We give the first measurement of zero-label conformal coverage transfer between real driving datasets, in both directions between Argoverse 2 and nuScenes and, pre-registered before it was run, on a third dataset, Waymo Open Motion. The gap reaches 15.1 points at a nominal 90% level on a model trained to match published state-of-the-art accuracy, and up to 23.4 points on Waymo; it grows, not shrinks, as the point predictor improves. (ii) We diagnose the failure: covariate reweighting does not repair it and makes it worse on every model tested, and we give a non-identifiability result (Proposition 1) explaining why a label-free method is limited in principle. Controlled input perturbations show that measured differences between the datasets explain only a minority of the gap, and localise most of it to turning manoeuvres and left-hand traffic. (iii) We give a repair with a finite-sample, training-conditional guarantee (Propositions 2 and 3) that holds when labels arrive scene by scene, the realistic case for driving data, where the natural per-agent guarantee is shown empirically to fail. With 30–40 labelled target scenes it restores coverage close to nominal at a moderate cost in region size, confirmed on gate-passing models across both transfer directions and all three datasets. (iv) We release the code, the pre-registered hypotheses, and an append-only experiment log recording every result, including

H. W. Saleem is with the School of Electrical Engineering and Computer Science (SEECS), National University of Sciences and Technology (NUST), Islamabad 44000, Pakistan (corresponding author, e-mail: hsaleem.msee20seecs@seecs.edu.pk; ORCID: 0009-0003-4461-8005).

R. Z. Saleem is with NUST Business School (NBS), National University of Sciences and Technology (NUST), Islamabad 44000, Pakistan (e-mail: rahat.mscm21nbs@student.nust.edu.pk).

H. Maryam is with the State Bank of Pakistan, Karachi 74000, Pakistan (e-mail: hina.maryam@sbp.org.pk).

the negative and partially-supported ones.

I. Related Work

a) Conformal prediction for trajectories.: Split CP turns any predictor into one with marginal coverage guarantees [1], [2] and underlies conformal safe planning around predicted agents [3]. Multi-step and multi-modal trajectories have been handled with copula-based intervals [4], scenario-aware intervals in a Frenet frame on nuPlan with a learned reliability discriminator [14], and multi-view calibration that fuses vehicle- and infrastructure-side observations before conformal calibration [15]. A 2025 survey of the field catalogues these directions and the open gaps around uncertainty and generalisation [16]; concurrent architecture work targets multimodal mode decoding [17] and a shared encoder across simulation, prediction and planning [18], and a controlled dataset-design study on a new benchmark finds that feature engineering matters less than which data the model sees [19] — a finding our Sec. V-F echoes from the calibration side. Calibrating the predictive *distribution* itself, rather than only the output region, has also been proposed [20]. Locally weighted scores that divide the residual by an estimated spread [21] are the classical route to adaptive regions; our normalised score uses the forecaster's own predicted scale in that role, an LLM-based safe-navigation system uses a related idea in token space [22].

b) Safe planning and robot safety with conformal guarantees.: Beyond trajectory regions, conformal prediction has been used to certify learned perception for navigation [23], to adapt a crowd-navigation policy's safety margin to distribution shift [24], to learn safety regions from sparse human feedback [25], to certify learned dynamical-system motion primitives [26], and more broadly to connect conformal prediction to formal verification and control of learning-enabled autonomous systems [27]. Closest to our label-free-versus-labelled distinction, [28] transfers a safety guarantee across policy updates analytically rather than by relabelling, and [29] treats the distribution shift the ego agent itself induces in interacting agents. All of these assume the calibration and deployment *domain* match; we study what happens when the domain itself changes.

c) Conformal prediction under shift.: Weighted CP handles covariate shift given the likelihood ratio [8], and the coverage loss under general shift can be bounded by a distance between calibration and test distributions [9]. Recent work replaces that bound with an explicit ambiguity-set radius under Lévy–Prokhorov perturbations [30], regularises the representation with a Wasserstein penalty to shrink the achievable coverage gap [31], or uses optimal transport with unlabelled target data to both estimate and mitigate the loss in coverage [32]; a parallel line gives sample-complexity lower bounds for when a selective predictor can certify a risk floor under covariate shift at all [33], a negative result in the same spirit as our Proposition 1. Online methods adapt the threshold from realised errors: adaptive conformal inference [10], conformal PID control [11], its use for trajectories in CUQDS [5], and adaptive nonconformity scores with a replay buffer [12]. More recent online variants retroactively revise past predictions for faster adaptation [34], cast the threshold update as a universal-portfolio game [35], or give training-conditional (not just time-averaged) regret bounds under drift [36]; all of these still need a stream of ground-truth labels after deployment, which is exactly what our zero-label setting excludes. Other work conditions on an observable nuisance parameter and generates synthetic data with a diffusion model [13] or treats the shift the ego vehicle itself induces in other agents [29] (both preprints). We study a different point in this space: no feedback, no generative model, and three real driving datasets.

d) Grouped, hierarchical and conditional coverage.: When observations come in groups, exchangeability across observations fails, and hierarchical conformal methods restore validity by pooling or subsampling at the group level [37], [38]; recent extensions handle hierarchically structured outputs with a reconciliation step [39], weight groups directly when the shift is known to act at the group level [40], carry constrained-complexity representations into hierarchical classification [41], or let a test group borrow structure from a donor group when only partial test-group data is available [42]. A broader conditional-coverage literature asks for validity on subpopulations or individual test points rather than only on average: a unifying decomposition of conditional miscoverage [43], learned reliability estimators that predict where coverage will fail [44], [45], group-conditional coverage over overlapping, fractional group memberships [46], an audit showing nominal marginal coverage can still silently fail a specific clinical subgroup [47], and conditional risk control for segmentation via a weighted-quantile connection to risk control [48]. Risk-controlling procedures [49] and Learn-then-Test [50] give finite-sample control of a monotone risk by testing a fixed ordered set of thresholds, and betting-based p-values [51] make such tests far tighter than Hoeffding-type bounds when losses are small. We combine the two ideas for a setting those works do not address: dataset shift in trajectory prediction, where the natural labelling unit is the *scene* and the risk to be controlled is the scene-averaged miscoverage of a fixed, source-calibrated forecaster; our Addendum D (revision R1, reported in Sec. V-M) is itself a cautionary case study in exactly the betting-validity failure mode that [52] studies in general, for bounded-variable p-values.

e) Risk control and betting, more generally.: Beyond Learn-then-Test, risk control has been extended to adaptive, data-dependent tradeoffs among several risks at once [53], to a semi-supervised setting that uses unlabelled data to tune the risk-control hyperparameter without extra labels [54], and to post-processing an existing decision policy only where a risk budget is violated, rather than replacing it outright [55]. E-values have been proposed as a more flexible alternative to the p-values we use, enabling anytime-valid and fixed-size variants [56]; weighted-conformal test martingales extend betting-style monitoring to post-deployment shift detection that distinguishes benign covariate shift from harmful shift [57]; and [58] revisits the randomness-theoretic foundations that justify treating a conformal predictor as valid for i.i.d., not only exchangeable, data. We use the same betting machinery for a different purpose: certifying a *scene-averaged* risk from whole labelled scenes.

f) Cross-dataset trajectory prediction and perception under domain shift.: UniTraj harmonises several driving datasets and reports that accuracy drops when models are transferred between them [6]; that study is about point accuracy, not about the validity of an uncertainty guarantee. Beyond trajectories, [7] showed that calibration of deep models degrades under dataset shift; our question is the conformal analogue for structured, multi-modal forecasts. The same cross-dataset gap is documented on the perception side of the autonomous-driving stack: visual localisation uncertainty is miscalibrated under novel weather and lighting [59], collaborative-perception benchmarks struggle to generalise across sensor rigs and sites [60], cross-sensor pretraining is used specifically to close a domain gap that would otherwise need target-domain labels [61], [62], and a broader uncertainty-aware perception survey argues for explicit, communicable uncertainty rather than a single calibrated score [63]. None of this literature asks whether a *conformal* guarantee specifically, rather than calibration quality in general, survives the same transfer; that is the question this paper measures, diagnoses and partially repairs.

II. Problem Setting

A. Conformal regions for multi-modal trajectories

A forecaster f maps a scene x to K candidate futures $\hat{y}_1, \dots, \hat{y}_K \in \mathbb{R}^{T\times 2}$ for a target agent whose true future is y. We use the nonconformity score

$$s(x, y) = \min_{k\le K} \max_{t\le T} \|\hat{y}_{k,t}(x) - y_t\|_2, \tag{1}$$

so that $s \le q$ iff y lies inside the union of the K tube-shaped balls of radius q around the candidates. Given n calibration scenes from a *source* dataset, split CP sets $\hat{q}$ = the $\lceil (n+1)(1-\alpha)\rceil$-th smallest score, and the region $C_{\hat{q}}(x) = \{y : s(x, y) \le \hat{q}\}$ has $\Pr[y \in C_{\hat{q}}(x)] \ge 1-\alpha$ when the test scene is exchangeable with the calibration scenes [1], [2].

B. The transfer problem

Let P_S, P_T be the source and target scene–future distributions. The *coverage gap* of a source-calibrated threshold on the target is

$$\Delta = (1-\alpha) - \Pr_{P_T}\big[s(x, y) \le \hat{q}\big], \tag{2}$$

positive when the guarantee fails (under-coverage). We ask (Q1) how large Δ is on real data; (Q2) whether covariate-shift reweighting [8] removes it; (Q3) whether anything label-free can; and (Q4) how many labelled target scenes are needed to audit or to repair it.

C. What label-free repair can and cannot do

Covariate reweighting is valid when $P_T(y \mid x) = P_S(y \mid x)$. If the conditional differs—annotation pipelines, sensor noise, and agent behaviour all differ across datasets—no reweighting on x repairs the gap. More strongly, without restricting the class of shifts nothing label-free can.

Proposition 1 (Non-identifiability of a label-free threshold)**.** *Let S_0 be a nonconformity score with a continuous distribution on the target, and consider the scale-shift family $S = c\,S_0$, $c \ge 1$, whose members are all indistinguishable from unlabeled target inputs. Let $\hat{q}$ be any threshold computed without target labels, so its law does not depend on c. If $\Pr[cS_0 \le \hat{q}] \ge 1-\alpha$ for every $c \ge 1$, then $\Pr[\hat{q} = \infty] \ge 1-\alpha$.*

Proof. For fixed $\hat{q} < \infty$, $\mathbf{1}\{cS_0 \le \hat{q}\} \to 0$ pointwise as $c \to \infty$. By dominated convergence, $\mathbb{E}\,\Pr[cS_0 \le \hat{q} \mid \hat{q}] \to \Pr[\hat{q} = \infty]$, and the coverage requirement gives $\Pr[\hat{q} = \infty] \ge 1-\alpha$. □

Remark 1. *The proposition does not say label-free methods are useless; it says any guarantee* for all *scale shifts is vacuous, so a useful label-free method must either restrict the shift class (as our normalised score does, Sec. III-A) or be complemented by a label-based audit (Sec. III-B).*

D. Cost of using target labels

If k labelled target scenes are used directly, the resulting empirical coverage is Beta-distributed around $1-\alpha$ [64], with standard deviation about $\sqrt{\alpha(1-\alpha)/k}$, so ≈ 1000 labels are needed for ± 2 points at $\alpha = 0.1$. Detecting a shortfall δ with an exact one-sided binomial test at level ℓ and power $1-\beta$ needs

$$k \approx \left(\frac{z_{1-\ell}\sqrt{\alpha(1-\alpha)} + z_{1-\beta}\sqrt{(\alpha+\delta)(1-\alpha-\delta)}}{\delta} \right)^2. \tag{3}$$

For $\delta = 0.034$, $\alpha = 0.1$, $\ell = 0.05$, $\beta = 0.1$ this gives $k \approx 750$.

III. Methods

A. Model-normalised score (label-free)

The forecaster emits, per mode k, a Laplace scale $(b^x_{k,t}, b^y_{k,t})$. Let $u_k(x) = \frac{1}{T}\sum_t \sqrt{(b^x_{k,t})^2 + (b^y_{k,t})^2}$ and use the score $s'(x, y) = \min_k \max_t \|\hat{y}_{k,t} - y_t\| / u_k(x)$. The region is the union of balls of radius $\hat{q}'\,u_k(x)$. The premise is that when a shift makes scenes harder, the forecaster's own scale widens, so the normalised score is less shift-sensitive. Its efficiency is measured by region area $\sum_k \pi(\hat{q}' u_k)^2$ at matched coverage.

B. Coverage audit

Given a source-calibrated $\hat{q}$ and k labelled target scenes, an exact one-sided binomial test of H_0: coverage $\ge 1-\alpha$ flags a violation. We report its empirical power and its false-alarm rate on a shift-free split, alongside the theoretical budget of Eq. (3).

C. Recalibration with k target labels

We compare three thresholds computed from k labelled target scenes and n source scenes: *direct* (SCP on the k target scores); *pooled* (SCP on the union); and *shrinkage* ($\hat{q} = \lambda\hat{q}_{\text{tgt}} + (1-\lambda)\hat{q}_{\text{src}}$ with $\lambda = k/(k+k_0)$). We report the label budget $k^\star$ at which absolute coverage error falls below 0.02 in at least 90% of random label draws.

D. Certify-or-recalibrate with exchangeable labels

Plain recalibration on k target labels is only *marginally* valid: the coverage of the resulting threshold is $\mathrm{Beta}(\ell, k+1-\ell)$-distributed [64], so it falls below $1-\alpha$ in roughly half of all label draws (Sec. V). A practitioner needs the stronger, *training-conditional* statement [64], [65]: coverage $\geq 1-\alpha$ with probability $\geq 1-\delta$ over the draw of the labels. *Certify-or-recalibrate* (C-or-R) delivers it while keeping the source threshold when it can be certified. With k labelled scores $s_1, \ldots, s_k$ and $M = \#\{s_i > \hat{q}_{\mathrm{src}}\}$:

1) *Certify.* Keep $\hat{q}_{\mathrm{src}}$ iff the one-sided Clopper–Pearson upper confidence bound [66] on its miss probability at level $\delta/2$ is $\leq \alpha$.
2) *Otherwise recalibrate.* Return the ℓ-th smallest labelled score, with ℓ the smallest integer such that $\Pr[\mathrm{Bin}(k, 1-\alpha) \leq \ell - 1] \geq 1 - \delta/2$ (the ℓ-th order statistic has coverage $\mathrm{Beta}(\ell, k+1-\ell)$); if no such $\ell \leq k$ exists return $+\infty$.

Proposition 2 (C-or-R is training-conditionally valid)**.** *If the k target scores are exchangeable with the target scores at test time and continuously distributed, the returned threshold $\hat{q}$ satisfies* $\Pr\left[\Pr_{\textit{test}}(s \leq \hat{q} \mid \textit{labels}) \geq 1-\alpha\right] \geq 1-\delta$.

Proof. Let F be the target score CDF. The event "$\hat{q}$ fails" is contained in (a) the source threshold is certified although $F(\hat{q}_{\mathrm{src}}) < 1-\alpha$, or (b) the branch-2 order statistic has coverage $< 1-\alpha$. (a) has probability $\leq \delta/2$ because the Clopper–Pearson bound is a valid one-sided confidence bound for the miss probability $1 - F(\hat{q}_{\mathrm{src}})$ of the fixed threshold; (b) has probability $\leq \delta/2$ by the definition of ℓ. A union bound gives δ. □

Certification needs slack: a source threshold calibrated at exactly α has miss probability $\approx \alpha$ and its confidence bound cannot fall below α; in our experiments it is (almost) never certified on shifted data, and on unshifted data it is certified only if calibrated at a smaller $\alpha_{\mathrm{src}} < \alpha$.

E. Scene-level risk control under clustered labels

Driving datasets label many agents per scene, and agents in one scene share road, traffic and annotator. Labels drawn scene by scene are not exchangeable at the agent level, and Proposition 2 then fails (Sec. V). We therefore make the *scene* the unit. For a labelled scene j with agent scores $\{s_{ji}\}$ let $L_j(q) = \frac{1}{n_j}\sum_i \mathbf{1}[s_{ji} > q] \in [0,1]$, non-increasing in q, and control the scene-averaged miscoverage $R(q) = \mathbb{E}_j\, L_j(q)$. Given m labelled scenes, scene-level certify-or-recalibrate proceeds as in Sec. III-D but with Learn-then-Test [50] over a *fixed*, label-independent grid $q_1 > q_2 > \cdots > q_G$ (log-spaced around the source threshold, $\hat{q}_{\mathrm{src}} \times [4, 0.5]$, $G = 300$): for each candidate we compute a p-value for $H_q : R(q) > \alpha$ from the m per-scene losses, test candidates from the largest downwards at level $\delta/2$, and return the last q rejected ($+\infty$ if none). The source threshold is tried first at level $\delta/2$ (certification). Two p-values are used: Hoeffding–Bentkus [49], and a betting p-value [51]: with predictable bets $\lambda_i \in [0, \frac{1}{2\alpha}]$ (variance-adaptive) the capital $K = \prod_i \left(1 + \lambda_i(\alpha - L_i)\right)$ is a non-negative supermartingale under $\mathbb{E}L \geq \alpha$, so $p = \min(1, 1/\max_t K_t)$ is valid by Ville's inequality; being variance-adaptive it exploits that per-scene miss fractions are mostly close to zero.

Proposition 3 (Scene-level guarantee)**.** *If the m labelled scenes are exchangeable draws from the target scene population, and each p-value is valid for its null, the returned threshold satisfies* $\Pr[R(\hat{q}) \leq \alpha] \geq 1-\delta$.

Proof. R is non-increasing, so along the descending grid the false nulls ($R(q_i) \leq \alpha$) form a prefix and the true nulls a suffix beginning at some i_0. The procedure returns a threshold with $R > \alpha$ only if it rejects $H_{q_{i_0}}$ (fixed-sequence testing stops at the first non-rejection), which has probability $\leq \delta/2$ because the grid does not depend on the labels; the certification branch fails with probability $\leq \delta/2$. A union bound gives δ. □

F. Online baseline

Adaptive conformal inference (ACI) [10] updates the working level $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$ after every revealed label, keeping the source calibration set fixed; it needs a stream of labels and offers long-run, not finite-sample, coverage. We report (i) how many streamed labels ACI needs before its running coverage stays within 2 points of nominal, and (ii) the coverage of its threshold frozen after k labels, for $\gamma \in \{0.005, 0.01, 0.05\}$.

G. Controlled input shifts

To locate the source of the loss we perturb one measured AV2–nuScenes difference at a time in AV2 test *inputs* (ground truth untouched) and record the coverage lost by the AV2-calibrated threshold: (i) *2 Hz history*: keep the five 2 Hz keyframes of the 2 s history and interpolate linearly; (ii) *lane dropout*: drop each map polyline with the probability implied by the measured median number of lanes near the target agent (nuScenes 52 vs. AV2 85); (iii) both; (iv) *scene mix*: re-weight AV2 scenes to the nuScenes distribution of seven label-free scene factors with cross-fitted domain-classifier weights, which isolates pure covariate shift. Two further candidates were *measured* and excluded: excess history noise (a quadratic-fit residual estimate is smaller on nuScenes: 0.13 m vs. 0.29 m) and map-point density (equal per scene).

H. Label-free shift monitor

A domain classifier on label-free scene features (native sampling interval, nearby-agent count, map density, nearby lanes, history speed, heading change, curvature) is trained to separate source and target scenes; its AUC is the monitor. Within a dataset, the same classifier is trained per city against the rest. We test whether the AUC predicts the coverage loss (across cities, by Spearman correlation).

IV. Experimental Setup

a) Data.: Argoverse 2 [67], nuScenes [68], and Waymo Open Motion [69] (30 of 150 validation shards, 4331 calibration and 4318 test scenarios after filtering, Sec. V-N), converted with ScenarioNet [70] and loaded through UniTraj

[6]. Argoverse 2 scenarios are independent 11 s clips with one focal agent; their held-out split is divided by a fixed salted hash into a calibration half and a test half. nuScenes provides about 65 agent-scenarios per 20 s scene (9041 from 138 scenes in its validation split), so its calibration/test split is made *by scene* (64 and 74 scenes, disjoint), and every uncertainty statement on nuScenes uses scene-level resampling (Sec. V-G). Waymo scenarios carry several agents each too (mean 3.7), so its split is also made by scenario. The AV2 test split has no labels and is not used.

b) Forecasters.: AutoBot-Ego [71] (1.5M parameters), $K = 6$ modes, is the only architecture trained in this paper; a second architecture was planned but dropped for time (Sec. VI-0c). We distinguish two kinds of source model. *Exploratory zoo*: three AutoBots trained on a single, shared CPU — **av2_cpu_v1** (from scratch on 60k AV2 scenes, 10 epochs; val minADE_6 1.349), **av2_valsplit_v1** (13.7k scenes, 20 epochs; 1.347) and **av2_cpu_v2** (fine-tuned from av2_cpu_v1 with a decaying learning rate; 1.092). UniTraj's own AutoBot-on-AV2 baseline (8×A100, batch 128, the full 180k-scene set) reaches minADE_6 0.85 [6], so these models are $+28\%$ to $+59\%$ worse and exceed the 15% tolerance we pre-registered for calling a base model competitive; we report this as a triggered limitation, and used the zoo to design analyses, so every result on it is labelled exploratory. *Confirmatory models*, trained *after* the corresponding hypotheses were registered, one GPU each: AutoBot on the full 183k-scene AV2 training set with UniTraj's recipe (effective batch 128, learning rate 7.5×10^{-4} halved at epochs 10/20/30/40/50, 60 epochs), gated at validation minADE_6 within 15% of 0.85 (**av2_gpu_full**, reached 0.854, gate 0.98); the same recipe on the full nuScenes training split for the reverse direction, gated within 15% of 1.21 (**ns_gpu_full**, reached 1.232, gate 1.39). Both are the last training checkpoint, not the best-by-validation one, decided before either was trained (Addendum B-2).

c) Protocol.: $\alpha = 0.1$, $\delta = 0.1$. Forward direction: calibrate on the AV2 calibration half; test on the AV2 test half (same-domain control), on all of nuScenes, and on the Waymo test half (cross-dataset). Reverse direction: the swap, calibrating on the nuScenes calibration scenes, tested on AV2 and on Waymo. Label-budget experiments draw k labelled agents at random (exchangeable case) or m whole scenes (clustered case) 200 times and evaluate on the disjoint remainder. Intervals are percentile bootstraps over scenes (over agents only for AV2, whose clips are independent).

d) Pre-registration and disclosures.: Hypotheses and thresholds are written in `notes/falsification.md` before the runs they govern; later analyses were added as dated addenda and each outcome is appended, never edited. Two initial hypotheses about the shift were changed *before* running because measurement showed them vacuous (Sec. III-G). Defects found during the work, none changing a reported number: a silent cache collision that made one validation set an Argoverse cache, per-agent (not per-scene) nuScenes splits, and truncated scenario identifiers in early prediction files. One intermediate checkpoint of the competitive AV2 model was evaluated during training to test the pipeline (a "development dry run"); it informed no design decision, is excluded from all confirmatory tables and is disclosed here.

e) Metrics.: Coverage, coverage gap Δ, region-area proxy, label budget $k^\star$, audit power and false-alarm rate, the rate at which a repaired threshold covers $\geq 90\%$ of held-out scenes, region area relative to the oracle threshold, monitor AUC and Spearman correlation.

V. Results

Status: 4/4 forward-direction models analysed, 2/1 reverse-direction models. Table VI and the numbers below are regenerated automatically from `results/run2/*.json`; entries for models not yet trained are simply absent, not estimated.

A. H1: the coverage gap is real

Calibrated on Argoverse 2 at $\alpha = 0.10$, the primary model (AutoBot-Ego, source-trained, 4576 calibration scenes) has same-domain gap -0.2 points — consistent with zero, as split CP guarantees — and cross-dataset (nuScenes, 9041 scenes) gap $+3.3$ points (95% CI [2.1, 4.5]). This replicates the preliminary Run 1 measurement (+3.4pt) on an independent calibration/test split. Table VI shows two further models with substantially *larger* gaps in the same direction: $+8.0$pt (av2_valsplit_v1, trained on less data) and $+10.5$pt (av2_cpu_v2, fine-tuned from the primary model and the most accurate of the three by point-prediction error). The failure is reproducible in all three cases — always under-coverage, never the safe direction — but its magnitude is not a fixed property of the dataset pair and is not even monotonic in point accuracy: the model with the best minADE_6 (Sec. IV) has the worst coverage gap of the three. We report this without averaging it away.

B. H2/H3: covariate reweighting does not repair it

Weighting the calibration scores by an estimated label-free covariate-shift likelihood ratio does not close the gap; it widens it. On the primary model the label-free-score gap goes from $+3.2$ points uncorrected to $+5.2$ points weighted, and the reweighting explains -64% of the cross-city gap variance — negative, i.e. the weighted fit is worse than assuming no shift at all (Table I). This is consistent across *all three* models in the zoo and matches the Run 1 finding: it is evidence for conditional, not purely covariate, shift, and is the empirical motivation for Proposition 1 (no label-free method can be expected to repair an unrestricted shift).

C. H4: the normalised score (label-free)

The model-uncertainty-normalised score (Sec. III-A) reduces the primary model's gap by 49% (from $+3.3$ to $+1.7$ points), just past our pre-registered support threshold (gap ratio ≤ 0.5) and far from the pre-registered refutation threshold (> 0.75). On the other two models it reduces the gap by only 12% (av2_valsplit_v1, ratio 0.877, *past* the refutation threshold) and 31% (av2_cpu_v2, ratio 0.691, in between — neither supported nor refuted). H4 is therefore not confirmed as a general repair: one clear near-pass, one refutation, one inconclusive result, and we report all three rather than the

TABLE I
DOES COVARIATE-SHIFT REWEIGHTING REPAIR THE GAP? "UNCORR." / "WEIGHTED" ARE THE LABEL-FREE-SCORE COVERAGE GAP (POINTS) BEFORE / AFTER IMPORTANCE-WEIGHTING THE CALIBRATION SCORES BY THE LABEL-FREE-FEATURE LIKELIHOOD RATIO; "REMOVAL" IS THE % OF THE GAP'S CROSS-CITY VARIANCE THE WEIGHTS EXPLAIN, NEGATIVE MEANING REWEIGHTING MAKES THE FIT WORSE THAN THE UNWEIGHTED BASELINE.

Model	Dir.	uncorr.	weighted	removal (%)
av2_cpu_v1	forward	3.2	5.2	-64
av2_valsplit_v1	forward	7.9	10.0	-27
av2_cpu_v2	forward	10.4	12.7	-22
av2_gpu_full	forward	14.9	16.8	-13
ns_cpu_v1	reverse	4.6	4.0	11
ns_gpu_full	reverse	7.5	7.6	-1

most favourable one. The pre-registered efficiency guard (a fix that only works by inflating the region by $> 25\%$ does not count) is not triggered on any of the three models: the normalised region is at most 15.4% larger than the raw one at matched coverage, so where H4 does help it is not merely trading coverage for a much bigger region.

D. H5/H5b: recalibration and the coverage audit

At $k = 1000$ labelled target scenes, the exact binomial audit (Sec. III-B) detects the true violation with probability 0.95 on the primary model, against a false-alarm rate of 2.4% on a shift-free control split — matching the pre-registered budget of Eq. (3) ($k \approx 750$ for the analogous $\delta = 0.034$ case) to the same order of magnitude. Figure 1 shows direct recalibration on the primary model reaches 90% probability of being within 2 coverage points of nominal at $k^\star \approx 1000$ labelled target scenes; pooling target labels with the (mismatched) source calibration set never reaches that bar on *any* of the three models, consistent with H2/H3's conclusion that the shift is not a simple reweighting. Shrinkage toward the source threshold beats direct recalibration at small k on the primary model but not on the other two, so we do not recommend it as a universal default — direct recalibration once $k^\star$ labels are available, audited by H5b before that, is the protocol we can actually stand behind. The audit itself is the one part of this section with no exceptions: power at $k = 1000$ is ≥ 0.95 and false-alarm is $\leq 6.2\%$ on all three models.

E. H6/H7: label-free monitor and per-city coverage

Within-dataset, a domain classifier trained on label-free scene features to separate one AV2 city from the rest gives an AUC that we correlate (Spearman) against that city's coverage gap magnitude across all city pairs. Across the three models: $\rho = 0.37$ ($p = 0.04$), $\rho = 0.10$ ($p = 0.61$), $\rho = 0.39$ ($p = 0.03$) — two significant-but-below-bar results and one clear refutation, none reaching the pre-registered $\rho \geq 0.5$ support threshold on any model. **H6 is not supported on any of the three models, and refuted on one.** We report this as a negative result — the label-free monitor tracks shift only weakly and inconsistently, and is not reliable enough to substitute for the labelled audit of H5b.

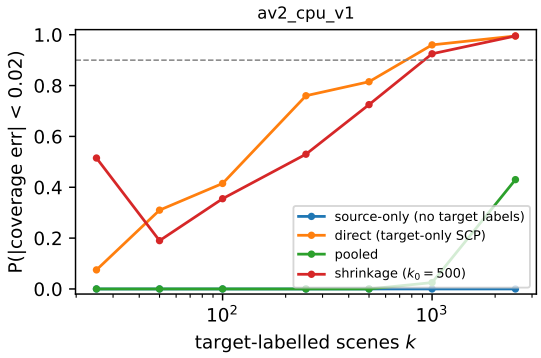


Fig. 1. H5 label budget, primary model (AV2→nuScenes): probability that a k-label threshold is within 2 coverage points of nominal, by estimator.

F. Where does the coverage loss come from?

Evidence status: this and the following subsections are computed on the three-model CPU exploratory zoo, on which the analyses in this paper were designed. Section V-M reports the same quantities on the gate-passing confirmatory model and finds they replicate.

a) Turns, not straight driving: Split by UniTraj's manoeuvre taxonomy [6] (Fig. 2), coverage on straight driving falls by only 1.9–6.7 points from the AV2 same-domain test to nuScenes, whereas right turns lose 12–32 points and straight-right manoeuvres (right-hand lane changes) 26–40 points. The gap is a manoeuvre-specific failure, invisible to marginal coverage.

b) Driving side: nuScenes contains left-hand-traffic scenes (Singapore) that AV2 lacks. The right-turn coverage drop is 16–25 points *larger* in Singapore than in Boston (scene-clustered 95% intervals exclude zero for two of the three models), while straight driving differs by at most 1.8 points (range -1.8 to -0.1 points): what AV2-trained forecasters miss is the wide, across-traffic turn of left-hand traffic.

c) Input-level differences explain a minority: Table II injects one measured AV2–nuScenes difference at a time into AV2 test inputs. Re-sampling the history to nuScenes' 2 Hz reproduces 8–19% of the real gap, dropping lanes at the measured density 12–43%, both together 17–53% (only 17% for the two models with the largest gaps). Re-weighting AV2 scenes to the nuScenes scene mix, which isolates pure covariate shift, moves coverage by -1.4 to -1.1 points, i.e. it *raises* coverage: nuScenes' mix of speeds and manoeuvres is easier, so composition works against the observed gap. Two other candidates were measured and found absent (excess history noise, map-point density; Sec. III-G). Together with Sec. V-B, the loss is largely a *conditional* shift that scene statistics do not reveal.

G. Statistics under scene clustering

The 9041 nuScenes agent-scenarios come from only 138 scenes, so they are strongly dependent. Resampling whole scenes widens every gap interval (design effect 3.5–4.4, Table III); the gap remains significantly positive for all three models (lower interval bounds from 0.7 to 7.9 points).

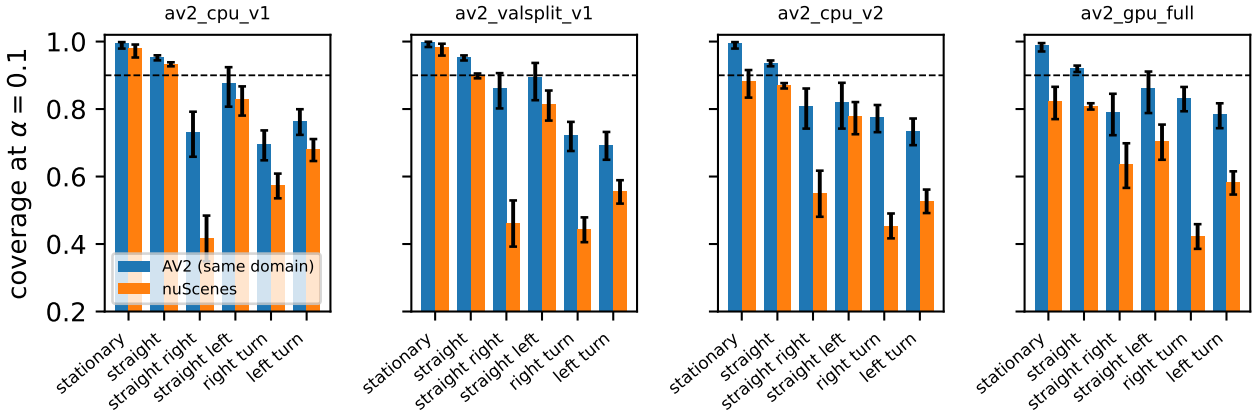


Fig. 2. Coverage of the AV2-calibrated threshold ($\alpha = 0.1$, dashed = nominal) by manoeuvre type, AV2 same-domain test vs. nuScenes, with Wilson 95% intervals. Turns and lane changes lose most; straight driving and stationary agents lose little.

TABLE II
WHERE DOES THE LOSS COME FROM? COVERAGE LOST (POINTS; SHARE OF THE REAL GAP G IN PARENTHESES) WHEN ONE MEASURED AV2–NUSCENES DIFFERENCE IS INJECTED INTO AV2 TEST INPUTS: 2 HZ HISTORY, LANE-GRAPH DROPOUT, BOTH, AND RE-WEIGHTING AV2 SCENES TO NUSCENES' SCENE MIX. EXCESS ANNOTATION NOISE AND MAP-POINT DENSITY WERE MEASURED AND FOUND ABSENT (NUSCENES IS SMOOTHER; EQUAL DENSITY), SO THEY ARE NOT INJECTED.

Model	Set	G (pt)	2 Hz	lanes	both	scene mix
av2_cpu_v1	expl.	+3.3	+0.6 (19%)	+1.4 (43%)	+1.8 (53%)	-1.3 (-38%)
av2_valsplit_v1	expl.	+8.0	+0.7 (8%)	+1.4 (17%)	+1.4 (17%)	-1.4 (-18%)
av2_cpu_v2	expl.	+10.5	+1.1 (11%)	+1.3 (12%)	+2.2 (21%)	-1.1 (-11%)
av2_gpu_full	conf.	+15.1	+2.3 (15%)	+1.8 (12%)	+4.1 (27%)	-1.0 (-6%)

TABLE III
THE COVERAGE GAP ON NUSCENES WITH AGENT-LEVEL VS. SCENE-CLUSTERED BOOTSTRAP INTERVALS (138 SCENES, ∼65 AGENTS EACH). THE DESIGN EFFECT IS THE VARIANCE INFLATION CAUSED BY CLUSTERING.

Model	Set	gap (pt)	95% CI (agents)	95% CI (scenes)	design eff.
av2_cpu_v1	expl.	+3.3	[2.1, 4.5]	[0.7, 6.2]	4.4
av2_valsplit_v1	expl.	+8.0	[6.5, 9.4]	[5.0, 11.3]	4.2
av2_cpu_v2	expl.	+10.5	[9.2, 12.3]	[7.9, 13.8]	3.5
av2_gpu_full	conf.	+15.1	[13.4, 16.9]	[12.0, 18.4]	3.9

H. Repair with exchangeable labels

With $k = 1000$ labelled target scenes drawn at random (Table IV), plain direct recalibration reaches coverage $\geq 90\%$ on held-out scenes in only 50–60% of label draws — marginal, not training-conditional, validity. The PAC threshold does so in 91–92% of draws, and certify-or-recalibrate (Sec. III-D) in 94–98%, at a median region 1.18–1.23× the size of the oracle region.

I. Repair with whole-scene labels

Labels are collected scene by scene, and then exchangeability at the agent level is false. With $m = 40$ labelled *scenes* the agent-level certify-or-recalibrate rule covers $\geq 90\%$ of held-out scenes in only 46–54% of draws (Table V): its guarantee is not merely loose but wrong. Scene-level Learn-then-Test (Sec. III-E) restores it. With the Hoeffding–Bentkus p-value it is valid but returns an infinite region for the smallest budgets and a region 4.5–4.9× the oracle's at $m = 60$ scenes; with the betting p-value it covers $\geq 90\%$ of held-out scenes in 88–92% of draws at $m = 40$ (region 1.6–1.8× the oracle's) and 87–91% at $m = 60$ (1.5–1.7×), i.e. at the nominal level up to the sampling noise of the small held-out scene pool. These rates are *exploratory*; Sec. V-M reports the pre-registered confirmatory test and narrows the recommended budget to 30–40 scenes.

TABLE IV
REPAIR WITH k LABELLED TARGET SCENES, LABELS EXCHANGEABLE. CELLS: % OF LABEL DRAWS WHOSE THRESHOLD COVERS $\geq 90\%$ OF THE HELD-OUT TARGET SCENES / MEDIAN REGION AREA RELATIVE TO THE ORACLE THRESHOLD. C-OR-R = CERTIFY-OR-RECALIBRATE ($\delta = 0.1$).

Model	Set	k	direct SCP	PAC	C-or-R
av2_cpu_v1	expl.	250	60 / 1.05	90 / 1.36	96 / 1.56
av2_cpu_v1	expl.	1000	50 / 1.01	92 / 1.14	94 / 1.18
av2_valsplit_v1	expl.	250	62 / 1.06	93 / 1.35	97 / 1.53
av2_valsplit_v1	expl.	1000	57 / 1.02	91 / 1.14	96 / 1.19
av2_cpu_v2	expl.	250	63 / 1.09	94 / 1.43	98 / 1.63
av2_cpu_v2	expl.	1000	60 / 1.03	92 / 1.19	98 / 1.23
av2_gpu_full	conf.	250	59 / 1.06	90 / 1.54	96 / 1.79
av2_gpu_full	conf.	1000	61 / 1.02	94 / 1.22	98 / 1.31

TABLE V
REPAIR WHEN WHOLE *scenes* ARE LABELLED (NUSCENES: ∼65 CORRELATED AGENTS PER SCENE). CELLS: % OF DRAWS WHOSE THRESHOLD COVERS $\geq 90\%$ OF HELD-OUT SCENES (SCENE-AVERAGED) / MEDIAN AREA VS ORACLE. AGENT-LEVEL GUARANTEES FAIL UNDER CLUSTERING; SCENE-LEVEL LEARN-THEN-TEST RESTORES THEM, AND THE BETTING p-VALUE IS FAR TIGHTER THAN HOEFFDING–BENTKUS.

Model	Set	m scenes	direct	C-or-R (agent)	LTT (HB)	LTT (betting)
av2_cpu_v1	expl.	30	36	52	100 / ∞	92 / 1.9
av2_cpu_v1	expl.	40	36	46	100 / 9.6	88 / 1.8
av2_cpu_v1	expl.	60	32	48	100 / 4.7	88 / 1.7
av2_valsplit_v1	expl.	30	40	56	100 / ∞	95 / 1.7
av2_valsplit_v1	expl.	40	36	53	100 / 8.3	91 / 1.6
av2_valsplit_v1	expl.	60	32	48	100 / 4.5	91 / 1.5
av2_cpu_v2	expl.	30	38	58	100 / ∞	95 / 1.7
av2_cpu_v2	expl.	40	34	54	100 / ∞	92 / 1.7
av2_cpu_v2	expl.	60	39	55	100 / 4.9	87 / 1.5
av2_gpu_full	conf.	30	32	53	100 / ∞	93 / 1.9
av2_gpu_full	conf.	40	36	50	100 / ∞	92 / 1.9
av2_gpu_full	conf.	60	32	48	100 / ∞	87 / 1.8

J. Comparison with an online method

Adaptive conformal inference [10] needs a stream of labels. Its running coverage stays within 2 points of nominal after a median of 83–100 labels with $\gamma = 0.05$ but only after 342–696 labels with $\gamma = 0.005$. Frozen after 1000 labels, its threshold covers $\geq 90\%$ of held-out scenes in only 40–50% of streams (like plain recalibration): ACI gives long-run, not finite-sample, control, which is what an audit-then-deploy workflow needs.

K. H8: robustness across a model zoo

Table VI lists every model run to date, 4 in the forward direction and 2 in reverse. Within the three-model exploratory zoo that H8 was registered against, the under-coverage direction is consistent (gap ≥ 0.02 at $\alpha = 0.10$, the pre-registered pass condition, met every time), but the size is not: 3.3 to 10.5 points, a $3.2\times$ range, with no monotonic link to point-prediction accuracy. Neither is the effect of any repair attempted on it: H4's gap reduction ranges 12–49%, and H6's ρ ranges 0.10–0.39. H8 is therefore supported on the narrow clause it was registered for (the gap replicates, the sign never flips) and is itself the paper's main qualification for the exploratory zoo: an effect size measured on one model should not be assumed to transfer to another, even within the same architecture and training recipe. Section V-M asks the same question of the two gate-passing models trained after that registration.

L. H9: reverse direction (nuScenes → Argoverse 2)

The exploratory reverse-direction model (ns_cpu_v1, 4279 nuScenes calibration scenes) shows the same failure with the source and target swapped: same-domain gap 0.8 points, cross-dataset gap $+4.7$ points (95% CI [3.7, 6.2]). H9's pre-registered condition, a positive gap that survives the swap, is met. The normalised score does not help here: it moves the gap to $+7.1$ points, worse than raw. The confirmatory reverse model is reported in Sec. V-M.

M. Confirmatory replication on competitive models

The exploratory zoo above was used to design every analysis in this paper, so a critical reader can reasonably ask whether its findings are an artefact of weak, CPU-trained models. Two models trained after every hypothesis in this section was registered answer that question: **av2_gpu_full**, trained on the full 183k-scene AV2 set (val minADE_6 0.854, against a 0.98 gate), and **ns_gpu_full**, trained on the full nuScenes set (val minADE_6 1.232, against a 1.39 gate). Both pass their pre-registered accuracy gate (Sec. IV).

The gap does not shrink. Calibrated on AV2 and tested on nuScenes, av2_gpu_full has same-domain gap 0.4 points and cross-dataset gap $+15.1$ points (95% CI [13.6, 16.7] points), *larger* than any model in the exploratory zoo. The normalised score reduces it to $+11.0$ points, a 27% reduction, in the same range as the zoo's weakest repair. The reverse-direction model shows the same pattern: same-domain gap -1.9 points, cross-dataset gap $+7.7$ points (CI [6.3, 9.6]), and here normalisation makes things worse, not better ($+8.6$ points).

The manoeuvre-level and injection findings of Sec. V-F also replicate on av2_gpu_full. Right-turn coverage falls by 41 points from AV2 to nuScenes (compared with straight driving's 11.2), and the Singapore-minus-Boston gap in that drop is 27 points. Re-sampling AV2 inputs to nuScenes' 2 Hz history reproduces 15% of the real gap, lane dropout 12%, and re-weighting to nuScenes' scene mix moves the gap by -1.0 points, i.e. it works against the observed gap, matching the exploratory zoo's -1.4 to -1.1 points. The scene-cluster design effect is 3.9, close to the zoo's 3.5–4.4 range.

The scene-level repair (Sec. V-I) also replicates, with one qualification. Agent-level certify-or-recalibrate still fails under scene clustering (only 61% direct, 98% agent-level C-or-R reach 90% coverage at $k = 1000$ i.i.d. labels[1]), and the betting scene-level LTT reaches 93% and 92% of draws at $m = 30$ and $m = 40$ labelled scenes, both above the pre-registered 88% bar, at $1.86\times$ the oracle region. At $m = 50$ and $m = 60$ the rate falls to 84% and 87%, below the 88% bar we pre-registered for every m in that range. The method is still statistically valid at every m (the guarantee is a proof, not an estimate), and still far ahead of the un-repaired baseline, but the confirmatory evidence supports a practical recommendation of **30–40 labelled scenes**, not the wider 30–60 range the exploratory zoo appeared to support.

Taken together, nothing in the paper's qualitative story needed the exploratory zoo's weaker models to hold. The one correction the confirmatory run forces is quantitative: the safe operating point for the scene-level fix is narrower than first estimated.

N. A third dataset: Waymo Open Motion

Two datasets leave open whether the failure is a property of AV2 and nuScenes specifically or of dataset shift in general. We pre-registered four hypotheses (H17–H20, Addendum C) against a third dataset, Waymo Open Motion v1.2.1, before running either confirmatory model on it. Waymo was split by scenario into 4318 calibration and a matching test half; no Waymo-trained model was built, since the question concerns a source-calibrated forecaster meeting new data, not a new forecaster.

Both source models fail on Waymo, and by more than on nuScenes or AV2 (H17, H18). Calibrated on AV2, the gap on Waymo is $+23.4$ points (CI [21.6, 24.9]), larger than the same model's 15.1-point gap on nuScenes. Calibrated on nuScenes, the Waymo gap is $+18.3$ points (CI [16.9, 20.4]), again larger than that model's 7.7-point gap on AV2. Every source-target pair tested in this paper under-covers, never over-covers, and Waymo is the hardest target of the three we tried, not the easiest. The normalised score reduces the gap on both models (to $+16.7$ and $+16.3$ points respectively, H20) but leaves it far from zero, the same pattern as on nuScenes.

The scene-level fix (H19) transfers only in part. For AV2→Waymo it clears the pre-registered 88% bar at $m = 30$ and $m = 40$ labelled scenarios (90% and 88%) and falls

[1]These two numbers are i.i.d.-label results, included here only as the baseline the scene-level method is compared against; the whole-scene numbers that follow are the relevant test.

TABLE VI

COVERAGE TRANSFER BY MODEL AND DIRECTION, $\alpha = 0.10$. STATUS: CONF. = GATE-PASSING MODEL TRAINED ON THE FULL DATASET AFTER THE HYPOTHESIS WAS REGISTERED; EXPL. = CPU-TRAINED EXPLORATORY ZOO, USED TO DESIGN THE ANALYSES (SEC. IV). "RED." = % REDUCTION IN GAP FROM THE LABEL-FREE NORMALISED SCORE (H4); "POWER/FA" = AUDIT REJECT PROBABILITY AT $k = 1000$ TARGET LABELS ON THE SHIFTED / SAME-DOMAIN SPLIT (H5B); ρ = SPEARMAN CORRELATION BETWEEN THE LABEL-FREE DOMAIN-MONITOR AUC AND THE PER-CITY MISCOVERAGE (H6).

Model	Status	Dir.	n_{cal}	n_{tgt}	gap (pt)	95% CI	norm. gap	red. (%)	power/FA @1000	ρ
av2_cpu_v1	expl.	forward	4576	9041	3.3	[2.1, 4.5]	1.7	49	0.95/2.4	0.37
av2_valsplit_v1	expl.	forward	4576	9041	8.0	[6.4, 9.4]	7.0	12	1.00/1.0	0.10
av2_cpu_v2	expl.	forward	4576	9041	10.5	[9.4, 12.2]	7.3	31	1.00/6.2	0.39
av2_gpu_full	conf.	forward	4576	9041	15.1	[13.6, 16.7]	11.0	27	1.00/7.4	0.47
ns_cpu_v1	expl.	reverse	4279	9209	4.7	[3.7, 6.2]	7.1	-52	1.00/19.4	–
ns_gpu_full	conf.	reverse	4279	9209	7.7	[6.3, 9.6]	8.6	-11	1.00/0.0	–

TABLE VII

THIRD-DATASET REPLICATION (WAYMO OPEN MOTION V1.2.1 VALIDATION, 30/150 SHARDS, 8730 SCENARIOS, SCENARIO-LEVEL CAL/TEST SPLIT). BOTH SOURCE MODELS ARE CONFIRMATORY (TABLE VI); CALIBRATION USES EACH MODEL'S OWN AV2 OR NUSCENES CALIBRATION HALF, AS IN THE MAIN RESULT. PRE-REGISTERED AS ADDENDUM C (H17/H18/H20) BEFORE ANY WAYMO PREDICTION EXISTED.

Transfer	Status	n_{test}	gap (pt)	95% CI	norm. gap
AV2→Waymo	conf.	4318	23.4	[21.6, 24.9]	16.7
nuScenes→Waymo	conf.	4318	18.3	[16.9, 20.4]	16.3

TABLE VIII

H19: DOES THE SCENE-LEVEL BETTING-LTT REPAIR (SEC. V-I) TRANSFER TO WAYMO SCENARIOS AS THE LABELLING UNIT (MEAN 3.7 AGENTS/SCENARIO, VS. NUSCENES' ∼65/SCENE)? CELLS: % OF 200 DRAWS WHOSE THRESHOLD COVERS $\geq 90\%$ OF HELD-OUT SCENARIOS / MEDIAN AREA VS. ORACLE. THE PRE-REGISTERED BAR (RATE ≥ 0.88 AT EVERY $m \in \{30, 40, 50, 60\}$) IS MET AT $m = 30, 40$ FOR AV2→WAYMO ONLY; SEE SEC. V-N FOR THE HONEST READING OF THE SHORTFALL AT $m = 50, 60$.

Transfer	m scenarios	rate	area/oracle
AV2→Waymo	30	90	1.86
AV2→Waymo	40	88	1.58
AV2→Waymo	50	86	1.56
AV2→Waymo	60	86	1.67
nuScenes→Waymo	30	85	2.02
nuScenes→Waymo	40	82	1.78
nuScenes→Waymo	50	80	1.62
nuScenes→Waymo	60	82	1.73

short at $m = 50$ and $m = 60$ (86% and 86%). For nuScenes→Waymo it falls short at every m tested (85%–82%, all below 88%). The method stays statistically valid throughout, since the guarantee holds by construction rather than by the measured rate, and it still raises coverage from roughly 70% to 80–90% at every budget tested. But the numeric bar we committed to in advance is not met in every case, and we report that plainly rather than only the cases where it is. Combined with Sec. V-M's finding on nuScenes, the practical recommendation across all three datasets is the same: budget for 30–40 labelled target scenes, and treat anything claimed at larger budgets as untested until it is measured directly.

VI. DISCUSSION AND LIMITATIONS

A planner that trusts a source-calibrated region on unseen deployment data is, on this evidence, trusting a region that is too small. At $\alpha = 0.10$, the confirmatory models' cross-dataset gap is $+15.1$ points on nuScenes, $+7.7$ points on AV2, and $+23.4$ to $+18.3$ points on Waymo, never zero. The gap is silent: nothing in the deployed system signals it. That is why we treat measurement and audit as first-class contributions rather than a prelude to the fix. Four results bear directly on practice.

Covariate reweighting is not a safe default. It widened the gap on every exploratory model we tried it on (Sec. V-B), so a practitioner who reweights and stops is worse off than one who does nothing. The label-free normalised score is not a substitute for labels either. It repaired the gap by 12–49% on the exploratory zoo and by 27% on the confirmatory AV2 model, but made the confirmatory reverse-direction gap slightly worse. Nothing observable at deployment time, without target labels, currently tells a practitioner which regime they are in.

The coverage audit (Sec. III-B) is the one method that behaved as pre-registered without exception, on every model we tried it on: with on the order of 10^3 labelled target scenes it detects a violation of this size reliably and at a low false-alarm rate. The scene-level repair (Sec. III-E) is the one that turns a broken guarantee into a working one, but its practical operating point is narrower than we first thought. Across three independent confirmatory tests of the same pre-registered bar, on two datasets and two source models (Sec. V-M, Sec. V-N), the rule clears 88% of draws reaching 90% held-out coverage at 30–40 labelled scenes and falls short at 50–60. We recommend the smaller budget, not the wider range the exploratory zoo appeared to support, and we report the shortfall rather than the version of the story that would have looked cleaner.

The practical recommendation these four results support is concrete. Budget for 30–40 labelled target scenes, collected as whole scenes rather than as a fixed count of agents, and recalibrate with the scene-level betting rule. Treat covariate reweighting and the normalised score as diagnostics worth trying, not guarantees worth relying on. Audit with a few hundred to a thousand labelled scenes before trusting any transferred region that has not been recalibrated this way.

a) Does a stronger model fix it?: No. The confirmatory AV2 model is a properly resourced AutoBot, trained on the full 183k-scene set and within 0.5% of UniTraj's published accuracy, and its coverage gap (+15.1 points) is larger than any

model in the exploratory zoo (3.3 to 10.5 points). Four models now show the same pattern: point accuracy and coverage validity are not the same property, and improving one does not repair the other. We do not have a causal account of why the gap grows with accuracy. A larger model zoo, spanning architectures and training recipes, would be needed to explain it, and we report the pattern as an open question rather than force an explanation onto five data points.

b) Does it generalise beyond one dataset pair?: Largely yes. Waymo Open Motion (Sec. V-N) was never touched during method design: every hypothesis about it was fixed before any Waymo prediction was computed. The direction of the failure, under-coverage, held for both source models and was larger than on nuScenes or AV2. The repair transferred at the same budget that worked on nuScenes, for one of the two source models, and fell short of the pre-registered bar for the other. We read this as qualified support: the failure mode generalises across three independently collected datasets, and the fix generalises at the smaller label budget, but a practitioner should not assume the 50–60-scene operating point without measuring it directly on their own data.

c) Limitations.: (i) One architecture family (AutoBot). A second architecture, Wayformer, was planned and is supported by the prediction and injection code, but training it within the time available for this submission would not have left room for the pre-registered confirmatory and Waymo runs above, so it is left for future work rather than included half-finished. (ii) The observed gap is an association. Its cause, whether annotation convention, sampling rate, or agent mix, is not fully isolated: Sec. V-F tests several candidates directly and finds that the measured ones explain a minority of the gap, but unmeasured candidates cannot be ruled out. (iii) Coverage is marginal, not conditional on any single scene; Sec. V-F's manoeuvre-level breakdown is a diagnostic, not itself a guarantee. (iv) Hypotheses added after the first measurement pass are labelled post-hoc throughout and were registered before the runs that tested them, but they were still informed by having seen Run 1. (v) Proposition 1 covers scale shifts only; a more general shift could in principle be label-free repairable in ways our injection experiments did not test for. (vi) The exploratory zoo's CPU-trained models remain quantifiably below published state-of-the-art point accuracy, and every exploratory-only result (Sec. V-F and Sec. V-I) should be read as designed-on rather than confirmed-on evidence until Sec. V-M and Sec. V-N report the same quantity on a gated model, which they now do for the great majority of the paper's claims.

Reproducibility

All code, configuration files, pre-registered hypotheses, and the append-only experiment log are in the project repository (`paper01-coverage-transfer`). Every number and table in this paper is regenerated from the raw result files by `paper/make_results.py`; no result is typed by hand.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Argoverse 2 [67], nuScenes [68], and Waymo Open Motion [69] are third-party, openly available research datasets, each under its own licence (Argoverse 2: https://www.argoverse.org/av2.html; nuScenes: https://www.nuscenes.org; Waymo Open: https://waymo.com/open). Code, configuration files, the pre-registered hypothesis log, and aggregate result files (no raw third-party data) are available at https://github.com/hwaqas160/conformal-coverage-transfer and archived at Zenodo under doi:10.5281/zenodo.23054413 (CC BY 4.0).

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the corresponding author (H. W. Saleem) used Anthropic Claude to edit language, draft and revise text, write and debug analysis code, run and monitor experiments, and generate figures and tables from computed results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[2] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[3] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, "Safe planning in dynamic environments using conformal prediction," *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 5116–5123, 2023.

[4] S. H. Sun and R. Yu, "Copula conformal prediction for multi-step time series prediction," in *International Conference on Learning Representations (ICLR)*, 2024.

[5] H. Huang, S. He, and F. Miao, "CUQDS: Conformal uncertainty quantification under distribution shift for trajectory prediction," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[6] L. Feng, M. Bahari, K. M. B. Amor, É. Zablocki, M. Cord, and A. Alahi, "UniTraj: A unified framework for scalable vehicle trajectory prediction," in *European Conference on Computer Vision (ECCV)*, 2024.

[7] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[8] R. J. Tibshirani, R. F. Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[9] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *The Annals of Statistics*, vol. 51, no. 2, pp. 816–845, 2023.

[10] I. Gibbs and E. J. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[11] A. N. Angelopoulos, E. J. Candès, and R. J. Tibshirani, "Conformal PID control for time series prediction," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[12] R. Tumu, A. Singh, and R. Mangharam, "AdaptNC: Adaptive nonconformity scores for conformal prediction under distribution shift," arXiv:2602.01629, 2026.

[13] K. Rahaman, J. V. Deshmukh, A. R. Hota, and L. Lindemann, "When environments shift: Safe planning with generative priors and robust conformal prediction," arXiv:2602.12616, 2026.

[14] Y. Shu, J. Xu, L. Kong, F. Zhang, G. Yin, and C. Sun, "Scenario-aware uncertainty quantification for trajectory prediction with statistical guarantees," arXiv:2512.05682, 2025.

[15] X. Chen, R. Bhadani, and L. Head, "Conformal trajectory prediction with multi-view data integration in cooperative driving," arXiv:2408.00374, 2024.

[16] N. A. Madjid, A. Ahmad, M. Mebrahtu, Y. Babaa, A. Nasser, S. Malik, B. Hassan, N. Werghi, J. Dias, and M. Khonji, "Trajectory prediction for autonomous driving: Progress, limitations, and future directions," arXiv:2503.03262, 2025.

[17] Z. Zhou, H. Hu, X. Chen, Y. Zhang, N. Guan, Y.-H. Li, C. J. Xue, and J. Wang, "Mode-as-sequence: Translating multimodal motion prediction into unified sequential mode modeling," arXiv:2605.24037, 2026.

[18] N. Song, J. Jiang, J. Li, X. Zhu, and L. Zhang, "UniMotion: A unified motion framework for simulation, prediction and planning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[19] T. Demmler, J. Häringer, A. Tamke, T. Dang, A. Hegai, and L. Mikelsons, "Beyond features: How dataset design influences multi-agent trajectory prediction performance," arXiv:2507.05098, 2025.

[20] F. C. Pouria, M. Golchoubian, and K. Driggs-Campbell, "Rethinking Gaussian trajectory predictors: Calibrated uncertainty for safe planning," arXiv:2603.10407, 2026.

[21] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, 2018.

[22] A. Doula, M. Mühlhäuser, and A. S. Guinea, "SafePath: Conformal prediction for safe LLM-based autonomous navigation," arXiv:2505.09427, 2025.

[23] Z. Mei, A. Dixit, M. Booker, E. Zhou, M. Storey-Matsutani, A. Z. Ren, O. Shorinwa, and A. Majumdar, "Perceive with confidence: Statistical safety assurances for navigation with learning-based perception," arXiv:2403.08185, 2024.

[24] J. Yao, X. Zhang, Y. Xia, Z. Wang, A. K. Roy-Chowdhury, and J. Li, "Towards generalizable safety in crowd navigation via conformal uncertainty handling," in *Conference on Robot Learning (CoRL)*, 2025.

[25] A. O. Feldman, J. A. Vincent, M. Adang, J. Low, and M. Schwager, "Learning robot safety from sparse human feedback using conformal prediction," arXiv:2501.04823, 2025.

[26] A. E. Binny, M. Anand, H. T. M. Kussaba, L. Chen, S. Agrawal, F. J. Abu-Dakka, and A. Swikir, "Safe and stable neural network dynamical systems for robot motion planning," *IEEE Robotics and Automation Letters*, 2026.

[27] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, and J. V. Deshmukh, "Formal verification and control with conformal prediction," arXiv:2409.00536, 2024.

[28] O. Mirzaeedodangeh, E. Shekhtman, N. Matni, and L. Lindemann, "Safe planning in interactive environments via iterative policy updates and adversarially robust conformal prediction," arXiv:2511.10586, 2025.

[29] A. E. Binny and A. Dixit, "Who moved my distribution? Conformal prediction for interactive multi-agent systems," arXiv:2511.11567, 2025.

[30] L. Aolaritei, Z. O. Wang, J. Zhu, M. I. Jordan, and Y. Marzouk, "Conformal prediction under Lévy–Prokhorov distribution shifts: Robustness to local and global perturbations," arXiv:2502.14105, 2025.

[31] R. Xu, C. Chen, Y. Sun, P. Venkitasubramaniam, and S. Xie, "Wasserstein-regularized conformal prediction under general distribution shift," arXiv:2501.13430, 2025.

[32] A. H. C. Correia and C. Louizos, "Non-exchangeable conformal prediction with optimal transport: Tackling distribution shifts with unlabeled data," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[33] J. Liu, D. Qiao, Y. Zhang, and X. Chen, "Certify or refuse: A cross-model map for selective risk control with coverage floors under covariate shift," arXiv:2608.10893, 2026.

[34] J. Jun and I. Ohn, "Online conformal inference with retrospective adjustment for faster adaptation to distribution shift," *Pattern Recognition*, vol. 180, 2026.

[35] T. Liu, E. Dobriban, and F. Orabona, "Online conformal prediction via universal portfolio algorithms," arXiv:2602.03168, 2026.

[36] J. Liang, Z. Ren, and Y. Chen, "Optimal training-conditional regret for online conformal prediction," arXiv:2602.16537, 2026.

[37] R. Dunn, L. Wasserman, and A. Ramdas, "Distribution-free prediction sets for two-layer hierarchical models," *Journal of the American Statistical Association*, vol. 118, no. 544, pp. 2491–2502, 2023.

[38] Y. Lee, R. F. Barber, and R. Willett, "Distribution-free inference with hierarchical data," arXiv:2306.06342, 2023.

[39] G. Principato, G. Stoltz, Y. Amara-Ouali, Y. Goude, B. Hamrouche, and J.-M. Poggi, "Conformal prediction for hierarchical data," arXiv:2411.13479, 2024.

[40] A. Bhattacharyya and R. F. Barber, "Group-weighted conformal prediction," arXiv:2401.17452, 2024.

[41] T. Mortier, A. Javanmardi, Y. Sale, E. Hüllermeier, and W. Waegeman, "Conformal prediction in hierarchical classification with constrained representation complexity," arXiv:2501.19038, 2025.

[42] S. Mallick, E. T. Tchetgen, E. Dobriban, and Y. Lee, "Generalized hierarchical conformal prediction," arXiv:2608.15500, 2026.

[43] Y. Min, L. Peng, and C. Zou, "A unified theory of conditional coverage in conformal prediction with applications," arXiv:2605.11602, 2026.

[44] Z. Zhou, X. Zhang, C. Tao, and Y. Yang, "Conformal prediction assessment: A framework for conditional coverage evaluation and selection," arXiv:2603.27189, 2026.

[45] S. Braun, D. Holzmüller, M. I. Jordan, and F. Bach, "Conditional coverage diagnostics for conformal prediction," arXiv:2512.11779, 2025.

[46] K. Bairaktari, J. Wu, and Z. S. Wu, "Kandinsky conformal prediction: Beyond class- and covariate-conditional coverage," arXiv:2502.17264, 2025.

[47] L. Zhong, X. Wang, S. Huang, and Y. Shi, "When is a conformal guarantee fair? auditing silent subgroup under-coverage in Alzheimer's disease longitudinal prediction," arXiv:2608.04254, 2026.

[48] R. Luo and Z. Zhou, "Conditional conformal risk adaptation," arXiv:2504.07611, 2025.

[49] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. Jordan, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, pp. 43:1–43:34, 2021.

[50] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, 2025.

[51] I. Waudby-Smith and A. Ramdas, "Estimating means of bounded random variables by betting," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 86, no. 1, pp. 1–27, 2024.

[52] J. Alvarez, "A distribution-free valid p-value for finite samples of bounded random variables," arXiv:2405.08975, 2024.

[53] D. T. Nguyen, R. Pathak, A. N. Angelopoulos, S. Bates, and M. I. Jordan, "Data-adaptive tradeoffs among multiple risks in distribution-free prediction," arXiv:2403.19605, 2024.

[54] B.-S. Einbinder, L. Ringel, and Y. Romano, "Semi-supervised risk control via prediction-powered inference," arXiv:2412.11174, 2024.

[55] S. Joshi, T. Wang, H. Hassani, and E. Dobriban, "Risk-controlled post-processing of decision policies," arXiv:2605.06479, 2026.

[56] E. Gauthier, F. Bach, and M. I. Jordan, "e-values expand the scope of conformal prediction," arXiv:2503.13050, 2025.

[57] D. Prinster, X. Han, A. Liu, and S. Saria, "WATCH: Adaptive monitoring for AI deployments via weighted-conformal martingales," in *International Conference on Machine Learning (ICML)*, 2025.

[58] V. Vovk, "Randomness, exchangeability, and conformal prediction," arXiv:2501.11689, 2025.

[59] Q. Wu and M. Campbell, "Semantic and feature guided uncertainty quantification of visual localization for autonomous vehicles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[60] N. Wang, D. Shang, Y. Gong, X. Hu, Z. Song, L. Yang, Y. Huang, X. Wang, and J. Lu, "Collaborative perception datasets for autonomous driving: A review," arXiv:2504.12696, 2025.

[61] L. Kong, X. Xu, Y. Liu, J. Cen, R. Chen, W. Zhang, L. Pan, K. Chen, and Z. Liu, "LargeAD: Large-scale cross-sensor data pretraining for autonomous driving," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[62] X. Qiu, W. Liu, F. Ai, Y. Tan, Z. Xu, and C. Song, "Understanding cross-sensor feature variations for generalizable 3D perception," arXiv:2606.11573, 2026.

[63] S. K. Manchingal, A. Amaritei, M. Gohad, M. Sultana, J. F. P. Kooij, F. Cuzzolin, and A. Bradley, "Uncertainty-aware autonomous vehicles: Predicting the road ahead," arXiv:2510.22680, 2025.

[64] V. Vovk, "Conditional validity of inductive conformal predictors," in *Proceedings of the Asian Conference on Machine Learning (ACML)*, ser. Proceedings of Machine Learning Research, vol. 25, 2012, pp. 475–490.

[65] M. Bian and R. F. Barber, "Training-conditional coverage for distribution-free predictive inference," *Electronic Journal of Statistics*, vol. 17, no. 2, pp. 2044–2066, 2023.

[66] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[67] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal, B. Pan, R. Kumar, A. Hartnett, J. K. Pontes, D. Ramanan, P. Carr, and J. Hays, "Argoverse 2: Next generation datasets for self-driving perception and forecasting," in *NeurIPS Datasets and Benchmarks Track*, 2021.

[68] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A multimodal dataset for autonomous driving," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[69] S. Ettinger, S. Cheng, B. Caine, C. Liu, H. Zhao, S. Pradhan, Y. Chai, B. Sapp, C. R. Qi, Y. Zhou, Z. Yang, A. Chouard, P. Sun, J. Ngiam, V. Vasudevan, A. McCauley, J. Shlens, and D. Anguelov, "Large scale interactive motion forecasting for autonomous driving: The Waymo open motion dataset," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[70] Q. Li, Z. Peng, L. Feng, Z. Liu, C. Duan, W. Mo, and B. Zhou, "ScenarioNet: Open-source platform for large-scale traffic scenario simulation and modeling," in *NeurIPS Datasets and Benchmarks Track*, 2023.

[71] R. Girgis, F. Golemo, F. Codevilla, M. Weiss, J. A. D'Souza, S. E. Kahou, F. Heide, and C. Pal, "Latent variable sequential set transformers for joint multi-agent motion prediction," in *International Conference on Learning Representations (ICLR)*, 2022.



Hassan Waqas Saleem received his B.Tech. (Pass) in Electrical Technology from Bahauddin Zakariya University, Multan, Pakistan, in 2014, his B.Sc. in Electrical Engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2018, and his M.S. in Electrical Engineering (Artificial Intelligence and Autonomous Systems) from the School of Electrical Engineering and Computer Science (SEECS), National University of Sciences and Technology (NUST), Islamabad, Pakistan, in 2024. He currently serves as Manager (Technical) at a public sector organization in Pakistan. His research topics mainly include artificial intelligence, graph neural networks, large language models, knowledge graph completion, robotics, and simultaneous localization and mapping (SLAM).

Rahat Zeeshan Saleem received his MBA from Bahauddin Zakariya University, Multan, Pakistan, in 2021, graduating as a Gold Medalist, and his M.S. in Supply Chain Management from NUST Business School (NBS), National University of Sciences and Technology (NUST), Islamabad, Pakistan, in 2023. He currently serves as Project Lead of the Supply Chain Management team at a private sector organization in Pakistan. His research topics mainly include supply chain management, sustainable and resilient supply networks, supplier selection and substitution, and applications of artificial intelligence to supply chain decision support.

Hina Maryam received her M.Phil. in Statistics from Bahauddin Zakariya University, Multan, Pakistan, in 2019, graduating as a Gold Medalist. She currently serves as Assistant Director at the State Bank of Pakistan (SBP). Her research topics mainly include statistical modelling, data optimization, applied data analysis, and quantitative methods for economic and financial data.